



The use of fenofibrate in diabetic retinopathy: Narrative review

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ABSTRACT

Diabetic retinopathy (DR) is a leading cause of blindness despite advances in treatment and there remains an unmet need for low-cost interventions to prevent and slow the progression of DR. Fenofibrate has shown promise as a useful adjunct in DR management and is licensed for use in the treatment of DR in a few countries. However, the data supporting the use of fenofibrate for the treatment of DR emerged from subsidiary analyses of the FIELD, ACCORD-LIPID and its sub-study ACCORD-EYE trials, which were primarily designed to evaluate cardiovascular outcomes. The recently concluded LENS trial, designed to evaluate the effect of fenofibrate on progression of DR, has shown clear evidence of benefit over medium-term follow-up (i.e., 4–5 years), and confirmed its safety in people with type 1 or type 2 diabetes. These benefits appear to be independent of fenofibrate's lipid lowering effects and are more likely to be mediated via its direct effects on the eye. In this narrative review on the use of fenofibrate in the treatment of DR, we summarise the possible mechanisms of action, existing evidence from randomised trials, and implications for clinical practice and research.

1. Background

Diabetic retinopathy (DR) is a leading cause of blindness worldwide, accounting for 2.6 % of cases of blindness and 1.9 % of cases of severe visual impairment.¹ It is the only major cause of blindness that has increased in prevalence from 1990 to 2020 in many regions of the world.² In 2020, 22.3 % of the global diabetes population had some form of DR.³ DR and related visual impairment significantly impair both general and vision-related quality of life.⁴ Longer diabetes duration and suboptimal glycaemic and blood pressure control are risk factors for DR.

Stringent glycaemic and blood pressure control are important to slow the progression of DR among people with recent onset diabetes.^{5–7} However, the benefit of these strategies is less marked among older adults with longstanding diabetes.^{8,9} National or regional retinal screening programmes exist to identify people with DR who may be at risk of vision-threatening disease, who may benefit from enhanced surveillance or ophthalmological treatment, and these programmes may also offer the opportunity to intervene at an earlier stage of disease if effective therapies can be identified. The current standard of care in DR

is largely aimed at treating people with vision-threatening DR (VTDR). For example, laser photocoagulation reduces the risk of severe vision loss by up to 50 % among people with proliferative DR (PDR).¹⁰ Focal laser photocoagulation reduces the rate of moderate visual loss (doubling of the visual angle / 15 or more letter loss on Early Treatment of Diabetic Retinopathy Study [ETDRS] visual acuity charts) from 24 % to 12 % at 3 years among people with clinically significant macular oedema (CSMO),^{10,11} and intravitreal administration of anti-vascular endothelial growth factor (anti-VEGF) drugs^{12–14} and steroids^{15,16} improve vision in people with clinically significant macular oedema (CSMO). However, these interventions are only offered to people with advanced disease, and options to slow progression of DR to these advanced stages are limited. Furthermore, laser therapy and intravitreal injections are not without limitations. Laser therapy can be associated with impaired light-dark adaptation, loss of peripheral vision, change in colour vision, and worsening of diabetic macular oedema (DMO).⁵ Intravitreal anti-VEGF injections are invasive, and they carry a risk of ocular infections, retinal detachment, vitreous haemorrhage, raised intraocular pressure and cataract formation.^{5,17}

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There is therefore a need for safe, cost-effective treatment options applicable across the spectrum of DR, beyond stringent glycaemic and blood pressure management. In this review, we appraise data from clinical trials suggestive of fenofibrate's DR-related benefit, summarise the possible mechanisms by which fenofibrate could confer retinal protective effects, discuss safety of fenofibrate therapy and implications for clinical practice and future research.

2. Benefits of fenofibrate in diabetic retinopathy

2.1. Clinical evidence

The FIELD¹⁸ and ACCORD-EYE¹⁹ trials were the first randomised clinical trials (RCTs) to demonstrate possible benefit of fenofibrate on DR-related outcomes in people with type 2 diabetes (T2D) (Table 1). Notably, both these trials were primarily designed to evaluate cardiovascular outcomes (no cardiovascular benefit was found) and DR-related endpoints were analysed as tertiary outcomes or pre-specified sub-studies. The MacuFen study²⁰ specifically investigated the role of fenofibrate among patients with early DMO (not requiring immediate laser or injection therapy), again in people with T2D. The recently published LENS trial is the first trial primarily designed to evaluate the effect of fenofibrate on DR progression in a population with early to moderate DR.²¹ Furthermore, it is the first to include people living with type 1 diabetes (T1D) as well as T2D. In this section, we summarise findings from hypothesis-generating clinic trials and randomised controlled trials designed to evaluate DR-related outcomes with fenofibrate therapy.

2.1.1. Hypothesis generating cardiovascular outcome trials

The FIELD study was a multicentre, placebo-controlled, double-masked, randomised clinical trial that investigated the cardiovascular effects of fenofibrate in adults living with T2D.²² A total of 9795 participants were randomised to receive micronised fenofibrate 200 mg daily or placebo and followed up for 2–5 years. The need for laser therapy for DR (a pre-specified tertiary outcome) was 31 % lower among participants assigned fenofibrate (164/4895) compared to participants assigned placebo (235/4900) (HR 0.69, 95 % CI 0.54–0.87, $p = 0.002$). The benefit was similar across both indications for laser therapy: PDR (HR 0.70, 95 % CI 0.52–0.93, $p = 0.015$) and DMO (HR 0.69, 95 % CI 0.54–0.87; $p = 0.002$).¹⁸ A retinal imaging sub-study of FIELD (the ophthalmology sub-study) included a subgroup of 1012 participants who underwent detailed retinal assessments. About 80 % of the sub-study participants did not have DR at baseline.¹⁸ There was no significant difference in DR progression (defined as worsening of DR grade by 2 or more steps on the ETDRS scale over a median follow up period of 5 years) between fenofibrate and placebo-treated groups (46/512 vs 57/500, $p = 0.19$).¹⁸ The investigators reported a significant reduction in DR progression in participants with DR at baseline (3.1 % in fenofibrate group vs 14.6 % in placebo group, $p = 0.004$), but no significant difference among those who did not have DR at baseline (11.4 % in fenofibrate group vs 11.7 % in placebo group; $p = 0.87$), with moderate evidence of a difference in the effect of fenofibrate therapy between these two groups (P value for interaction 0.019).

ACCORD was a multicentre, double-masked, placebo-controlled, randomised clinical trial with 10,251 participants living with T2D. Using a factorial design, it investigated the effect of strict glucose control, strict blood pressure control and, separately (in the ACCORD-LIPID study), treatment with fenofibrate vs. placebo (added to background simvastatin) on cardiovascular outcomes. The ACCORD-LIPID study included 5518 participants with moderate dyslipidaemia who were randomised to receive fenofibrate 160 mg daily or placebo. ACCORD-EYE was a sub-study that investigated the effect of the above interventions (simvastatin with fenofibrate vs simvastatin with placebo) on progression of DR. As part of this sub-study, retinal photographs were obtained at baseline and at 4 years of follow up. Among those who had

4-year outcome data (and who were therefore included in the analysis), 50.8 % had no DR at baseline.⁹ A total of 1918 participants in the ACCORD-EYE sub-study were part of the ACCORD-LIPID study and therefore were randomised to receive either fenofibrate ($n = 959$) or placebo ($n = 959$) (added to background simvastatin). At four years' follow up, data were available from 1593 individuals (806 in fenofibrate group and 787 in placebo group). A 40 % reduction in the sub-study's primary ophthalmic outcome was observed in ACCORD-EYE (namely the composite of the need for laser therapy or progression of DR by 3 or more steps on the ETDRS scale or vitrectomy: 52/806 in fenofibrate group vs 80/787 in placebo group, adjusted OR 0.60, 95 % CI 0.42–0.87, $p = 0.006$).⁹ There was a suggestion of greater benefit in individuals with DR at baseline (OR 0.45, 95 % CI 0.28–0.73, $p = 0.001$) compared to those without (OR 0.97, 95 % CI 0.55–1.73, $p = 0.92$) (p value for interaction 0.03).⁹

In a meta-analysis of FIELD ($n = 9795$, 5-year follow up data) and ACCORD-EYE ($n = 1918$, participants included in both ACCORD-EYE and ACCORD-LIPID, 4-year follow up data), fenofibrate decreased the need for focal or grid laser therapy (134/5854 in fenofibrate group vs 193/5859 in placebo group, OR 0.69, 95 % CI 0.56–0.86, $p = 0.0009$) and the need for pan-retinal photocoagulation (83/5854 in fenofibrate group vs 123/5859 in placebo group, OR 0.67, 95 % CI 0.51–0.89, $p = 0.005$).²³ However, among those without DR at enrolment, incidence of overt retinopathy was not different between individuals treated with fenofibrate or placebo (166/836 in the fenofibrate group vs 172/795 in placebo group, OR 0.91, 95 % CI 0.76–1.09, $p = 0.30$).²³ The lack of benefit in individuals without established DR may be due to the slow nature of disease progression, which could make these analyses under-powered (in terms of group sizes and durations of follow up) to detect a difference.

In another meta-analysis, Preiss and colleagues combined data from three cardiovascular outcome trials of fenofibrate - FIELD, ACCORD-LIPID and the Lipids in Diabetes Study^{18,19,24} ($n = 19,504$, weighted mean follow up 4.0 years) - and concluded that fenofibrate may decrease the need for laser therapy by about 25 % (391/9754 in fenofibrate group, 501/9750 in placebo group, OR 0.77, 95 % CI 0.67–0.88, $p = 0.0001$).²⁵ The difference in event rates was more apparent after the first year (223/9754 in fenofibrate vs 316/9750 in placebo group, OR 0.70, 95 % CI 0.58–0.83, $p < 0.0001$) than during the first year of treatment (168/9754 in fenofibrate group vs 185/9750 in the placebo group, OR 0.90, 95 % CI 0.73–1.12, $p = 0.35$).

It is noteworthy that in both FIELD and ACCORD-LIPID trials, changes in lipid parameters do not suggest blood lipid lowering to be the main driver of retinal benefits. Among FIELD study participants, there was no significant difference in baseline lipid parameters among participants who required laser therapy, compared to those who did not. Furthermore, the differences in blood lipids between fenofibrate group and placebo group attenuated over the course of the study while differences in retinopathy outcomes continued to widen. Among ACCORD-LIPID study participants, the differences in absolute reductions in atherogenic lipids were small (LDL Cholesterol reduction: from 2.59 mmol/L to 2.10 mmol/L in fenofibrate group, from 2.61 mmol/L to 2.07 mmol/L in placebo group).

FIELD and ACCORD-LIPID trials were the first large scale clinical trials that showed the promise of fenofibrate improving ocular outcomes in people living with T2D and established DR. Their findings led to fenofibrate being approved for use in the treatment of DR in some countries. However, the primary cardiovascular outcomes of both FIELD and ACCORD-LIPID trials demonstrated no cardiovascular benefit of fenofibrate therapy and it is possible that evidence of ocular benefit in analyses of subsidiary outcomes may have represented false positive results. There was, therefore, a need for randomised controlled trials primarily designed to investigate the effect of fenofibrate on DR progression.

Table 1

Randomised controlled trials assessing the role of fenofibrate in the treatment of diabetic retinopathy.

Study	Design	Population	Intervention and control	Outcome	Results of primary outcome	Other results
<i>Hypothesis generating cardiovascular outcome trials</i>						
FIELD ^{18,22}	Factorial design, multi-centre, placebo controlled randomised clinical trial	Adults with T2D with or without DR at baseline N = 9795 DR at baseline: 8.3 % (self-reported)	Intervention: Micronised fenofibrate 200 mg once a day (n = 4895) Control: placebo (n = 4900) Median follow up: 5 years	Primary: Coronary events (fatal or non-fatal MI) Ocular outcome(s): Need for laser treatment for DR (tertiary outcome)	Coronary events: no difference (HR 0.89, 95 % CI 0.75–1.05; p = 0.16) Need for laser therapy: Reduced (164/4895 [3.4 %] in fenofibrate group vs 238/4900 [4.9 %] in placebo group, HR 0.69, 95 % CI 0.54–0.87, p = 0.002)	
FIELD ophthalmology sub-study ^{18,22}	Sub-study of FIELD. Retinal imaging at baseline, 2 years, 5 years and trial end.	Adults with T2D with or without DR at baseline N = 1012 DR at baseline: 20 %	Intervention: Micronised fenofibrate 200 mg once a day (n = 512) Control: placebo (n = 500) Median FU: 5 years	Primary: Progression of DR by 2 or more steps in ETDRS	DR progression: no difference (4 [9.6 %] on fenofibrate vs 57 [12.3 %] on placebo; p = 0.19)	DR progression significantly lower among individuals with DR at baseline (3 [3.1 %] on fenofibrate vs 14 [14.6 %] on placebo; p = 0.004), but not among individuals without DR at baseline (4 [11.4 %] on fenofibrate vs 43 [11.7 %] on placebo, p = 0.87), test for interaction p = 0.019 Exploratory composite outcome of DR progression, macular oedema and need for laser: improved (HR 0.66, 95 % CI 0.47–0.94; p = 0.022). No difference in need for vitrectomy (2 [0.4 %] vs 1 [0.2 %], p = 0.73) No difference in worsening of visual acuity (by 2 or more lines in Snellen chart) (97 [30.7 %] vs 90 [29.1 %], p = 0.67)
ACCORD-EYE ^{19a}	Sub-study of ACCORD trial (a multi-centre, double masked, placebo controlled clinical trial with factorial design). Retinal imaging at baseline and at 4 years of follow up.	Adults with T2D with or without DR at baseline Among all randomised, 1918 individuals belonged to both ACCORD-EYE and ACCORD-lipid sub-studies. Follow-up data were available from 1593 participants.	Fenofibrate 160 mg/day (n = 806) vs placebo (n = 787) as add-on to simvastatin. FU assessment at 4 years	Primary: composite of progression in ETDRS scale by 3 or more steps or need for laser photocoagulation or vitrectomy	Improved composite primary outcome: 52/806 (6.5 %) with fenofibrate, vs 80/787 (10.2 %) with placebo, aOR 0.60, 95 % CI 0.42–0.87, p = 0.006. Benefit greater with DR at baseline: 27/405 (6.7 %) with fenofibrate vs 56/412 (13.6 %) with placebo, OR 0.45, 95 % CI 0.28–0.73, p = 0.001 but not among those without DR at baseline: 25/401 (6.2 %) with fenofibrate vs 24/375 (6.4 %) with placebo, OR 0.97, 95 % CI 0.55–1.73, p = 0.92. p value for interaction 0.03.	Other outcomes: Need for laser therapy: NS. Focal laser: 0.72 (0.39, 1.31) p = 0.28 Scatter laser: 0.51 (0.21, 1.21) p = 0.12 Moderate vision loss - NS (23.7 % vs 24.5 %, aHR 0.95, 95 % CI 0.79–1.14, p = 0.57)
<i>Trials designed to evaluate eye outcomes</i>						
MacuFen ²⁰	Multi-centre RCT	Adults with DMO and T2D, not requiring immediate laser or anti-VEGF therapy (N = 110)	Fenofibrate 135 mg/day (n = 57) vs placebo (n = 53). Follow up at 1 year.	Primary outcome: change in total macular volume (TMV)	No significant difference in change in TMV (between group difference: −0.25 mm ³ (95 % CI, −0.645, 0.155; p = 0.227))	No significant difference in change in ETDRS scale (worse eye analysis): Worsening by 2-steps: 8 % vs 14 % Improvement by 2-steps: 15 % vs 16 % Overall trend: p = 0.228 Change in VA: Between group difference: 0.02 (95 % CI −0.05, 0.09, p = 0.623)

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Table 1 (continued)

Study	Design	Population	Intervention and control	Outcome	Results of primary outcome	Other results
Srinivasan et al 2017 ²⁷	Single centre randomised, placebo controlled clinical trial	Adults with T2D and treatment naïve centre-involving DMO (central macular thickness [CMT] > 250 µm)	Intervention: fenofibrate 160 mg once a day (25 individuals, 28 eyes) Control: placebo (235 individuals, 25 eyes) Follow up – 2 months for 6 months	Co-primary outcome: change in central macular thickness	Significant reduction in CMT at 6 month follow up. Between group difference –25.196 µm, p = 0.031	No significant difference in visual acuity change. Between group difference –0.068, p = 0.186
LENS ²¹	Multi-centre, double-masked, placebo-controlled, randomised clinical trial	Adults with T1D or T2D and non-referable DR or maculopathy.	Intervention: Nanoparticle fenofibrate 145 mg once a day (n = 576) Control: Matched placebo (n = 575). Median FU 4.0 years Fenofibrate given on alternate days if eGFR 40–59 mL/min/1.73 m ²	Composite of developing referable DR or maculopathy or treatment for DR or maculopathy (intravitreal medication injection, retinal laser, vitrectomy)	Improved composite primary outcome (131/576 [22.7 %] vs 168/575 [29.2 %], HR 0.73; 95 % CI, 0.58–0.91; p = 0.006)	Any progression of DR: 185/576 [32.1 %] vs 231/575 [40.2 %], HR 0.74; 95 % CI, 0.61–0.90, p = 0.003 Progression to referable DR/ maculopathy: 130/576 (22.6 %) vs 168/575 (29.2 %), HR 0.72, 95 % CI 0.57–0.91, p = 0.005 Development of macular oedema: 22/576 [3.8 %] vs 43/575 [7.5 %], HR 0.50; 95 % CI, 0.30–0.84, p = 0.008 Need for treatment: 17/576 [3.0 %] vs 28/575 [4.9 %], HR 0.58, 95 % CI 0.31–1.06, p = 0.08 (laser therapy 1.7 % vs 3.0 %; intravitreal injections 1.6 % vs 2.1 %; vitrectomy 0.0 % vs 0.7 %)

aHR: adjusted hazard ratio, aOR: adjusted odds ratio, CI: confidence interval, CI-DMO: centre-involved diabetic macular oedema, CMT: central macular thickness, DMO: diabetic macular oedema, DR: diabetic retinopathy, eGFR: estimated glomerular filtration rate, ETDRS: Early Treatment of Diabetic Retinopathy Study, FU: follow up, HR: hazard ratio, NPDR: non-proliferative diabetic retinopathy, NS: non-significant; OR: odds ratio, PDR: proliferative diabetic retinopathy, RCT: randomised controlled trial, T1D: type 1 diabetes, T2D: type 2 diabetes, TMV: total macular volume, VA: visual acuity.

^a ACCORD-EYE (n = 3472, 50.8 % had no DR at baseline) and ACCORD-LIPID (n = 5518) were sub-studies of ACCORD trial (n = 10,251). 1918 participants were part of both ACCORD-EYE and ACCORD-LIPID sub-studies and among them 4-year follow up data were available from 1593.

2.1.2. Randomised controlled clinical trials designed to investigate the effect of fenofibrate on eye outcomes

The recently published LENS trial investigated the impact of fenofibrate on the progression of DR (Table 1).²¹ LENS was a multicentre, randomised, parallel-group, double-masked, placebo-controlled clinical trial that was embedded within routine clinical care in Scotland. It included adults with diabetes (either T1D or T2D) and non-referable DR (defined as mild background retinopathy or observable background retinopathy or observable maculopathy, according to NHS Scotland's Diabetic Eye Screening programme grading scheme). Eligible participants were randomised in a 1:1 ratio to receive either fenofibrate 145 mg nano-particle tablet once a day or matching placebo. This dose of fenofibrate nano-particle formulation is bioequivalent to micronised formulations used in FIELD and ACCORD-Eye studies.²⁶ The primary outcome was time to the first occurrence of the composite of either developing referable DR or maculopathy, or treatment for DR or maculopathy (including intravitreal injection of medication, retinal laser therapy, or vitrectomy) in either eye. The study included 1151 participants (26 % with T1D) who were followed up over a median duration of 4.0 years. The primary outcome was significantly lower in participants assigned fenofibrate compared to participants assigned placebo (131/576 [22.7 %] vs 168/575 [29.2 %], HR 0.73; 95 % CI, 0.58 to 0.91; $p = 0.006$) reflecting an absolute risk reduction of 6.5 percentage points (95 % CI, 1.4 to 11.5 percentage points). Reductions were also observed in secondary outcomes including any DR progression (by one or more steps) in Scotland's Diabetes Eye Screening grading scale (185/576 [32.1 %] vs 231/575 [40.2 %], HR 0.74; 95 % CI, 0.61–0.90), progression to referable maculopathy (107/576 [18.6 %] vs 149/575 [25.9 %], HR 0.66, 95 % CI, 0.52–0.85) and the development of macular oedema (22/576 [3.8 %] vs 43/575 [7.5 %], HR 0.50; 95 % CI, 0.30–0.84). A non-significant reduction in treatment for DR (i.e. laser photocoagulation, anti-VEGF injections or vitrectomy) (17/576 [3.0 %] vs 28/575 [4.9 %], HR 0.58, 95 % CI 0.31–1.06) was observed, though it should be noted that the trial was not powered to detect a reduction in DR treatment and that this result is consistent with the other eye outcomes. The observed impact of fenofibrate did not differ between individuals with T1D and T2D, or between individuals with estimated glomerular filtration rate (eGFR) less than or greater than 60 mL/min/1.73 m². The latter is notable given that those with eGFR of 30–60 mL/min/1.73 m² only took one study tablet (fenofibrate 145 mg or placebo) on alternate days during the trial due to the lack of a lower dose option. Furthermore, changes in atherogenic lipids were small (mean non-HDL cholesterol in fenofibrate group: from 105.8 mg/dL (95 % CI 103.0–108.7) to 103.3 mg/dL (95 % CI 101.3–105.3), in placebo group: from 106.2 mg/dL (95 % CI 103.1–109.3) to 106.3 mg/dL (95 % CI 104.3–108.3)) and considered unlikely to be driving the observed improvements in retinal outcomes.

The findings of the LENS trial are quantitatively similar to the effects observed in the FIELD and ACCORD-EYE studies, as well as the meta-analyses (see Section 2.1.1). However, FIELD and ACCORD-EYE recruited individuals with dyslipidaemia, whereas LENS did not consider baseline lipid status. The findings in LENS suggest that fenofibrate could be effective at reducing DR progression even in those with normal lipid profiles.

Two other smaller randomised clinical trials designed to investigate the effect of fenofibrate on DR (in this case, the treatment of DMO) have been reported. The MacuFen study explored the effect of fenofibrate, compared to placebo, in people with DMO who did not require immediate laser or anti-VEGF therapy.²⁰ Among 110 adults included in this study, 57 individuals (83 eligible eyes) received fenofibrate and 53 individuals (77 eligible eyes) received placebo. All participants were followed up for one year. No significant difference in the change in macular volume between the groups was found (between-group difference in total macular volume change was -0.25 mm^3 [95 % CI, $-0.645, 0.155$; $p = 0.227$]). Srinivasan and colleagues reported a prospective randomised study designed to investigate the effect of fenofibrate as an

adjunct to standard treatment for DMO. Fifty treatment-naïve individuals with DMO were randomised to receive fenofibrate 160 mg daily for 6 months in addition to standard DMO treatment or standard DMO treatment alone (anti-VEGF therapy and/or focal/grid laser therapy). Fenofibrate as an adjunct to standard DMO treatment led to a greater reduction in central macular thickness compared to standard DMO treatment alone (136 μm vs 83 μm , $p = 0.031$).²⁷ There was no significant difference in visual acuity. These two studies are limited by small sample size and short duration of follow-up.

Published randomised trials^{18,20,28,29} and a meta-analysis³⁰ have not shown a benefit of fenofibrate in improving or preserving visual acuity. This is likely a result of most participants in these studies having no or only early DR with well-preserved vision.

In summary, the FIELD and ACCORD-LIPID trials, and their sub-studies that evaluated the effect of fenofibrate on improving DR as a tertiary outcome, were the first to report clinical evidence of a potential role of fenofibrate in the management of DR. The recently concluded LENS trial investigated the use of fenofibrate in reducing DR progression among patients with non-referable DR as its primary outcome and has confirmed this benefit. Sustainability of these beneficial effects during longer term treatment with fenofibrate remains to be seen. Data on preservation of visual acuity, as well as the role of fenofibrate in patients with established vision-threatening DR, are limited.

2.2. Mechanism of action

Fenofibrate is a prodrug. It is cleaved by tissue and plasma esterases to produce active fenofibric acid. It is an activator of the nuclear receptor peroxisome proliferator activator receptor- α (PPAR α). Its main use is in the treatment of hypertriglyceridaemia. However, the causal relationship between blood lipid levels and risk of developing DR is unclear.³¹ Increased serum triglyceride and decreased serum HDL-cholesterol concentrations are associated with increased prevalence and severity of DR.^{7,32} However, statin therapy has not shown consistent benefit in retarding progression of DR.³⁰ Furthermore, in the FIELD trial, baseline lipid profiles were not different between participants who needed retinal laser therapy and those who did not. The improvements in lipid profile with fenofibrate attenuated over the course of the study, while retinal benefits of fenofibrate continued to increase.¹⁸ Similarly, despite the clear difference in retinal benefits between fenofibrate and placebo-treated groups in LENS trial, their differences in lipid levels were small.²¹ Therefore, it is likely that beneficial effects of fenofibrate are independent of its blood lipid lowering effect.

PPAR α protects the retina against oxidative stress resulting from mitochondrial dysfunction and therefore, against retinal cell apoptosis.³³ In animal models, PPAR α upregulation inhibited reactive oxygen species-mediated apoptosis of retinal endothelial cells and pericytes,³⁴ thus protecting the outer blood retinal barrier.³⁵ PPAR α upregulation with intravitreal as well as oral administration of fenofibrate in mouse model with T1D leads to suppression of hypoxia-inducible factor-1 α (HIF-1 α) expression (HIF-1 α is a potent stimulant for vascular endothelial growth factor [VEGF] expression), VEGF expression (a key driver of PDR and DMO), endothelial cell migration, neovascularisation and vascular leakage.^{35–37} This suggests that local retinal effects of fenofibrate's PPAR α -dependent actions within the retina may drive its protective effects against DR progression (Fig. 1).^{35,38}

Fenofibrate may also have PPAR α -independent mechanisms of retinal protection. For example, it protects the blood-retinal barrier (BRB) against injury from modified low-density lipoprotein (LDL) particles.^{35,38} BRB is comprised of an inner BRB (endothelial cells and their tight junctions) and an outer BRB (retinal pigment epithelial cells and their tight junctions). An intact BRB prevents lipid extravasation into the retina, independent of blood lipid levels.³⁹ However, BRB damage could lead to extravasation of LDL particles into the retina, and these could then be modified to form oxidised LDL or glycated LDL (i.e., modified LDL).^{38,39} Modified LDL particles can cause further disruption of the

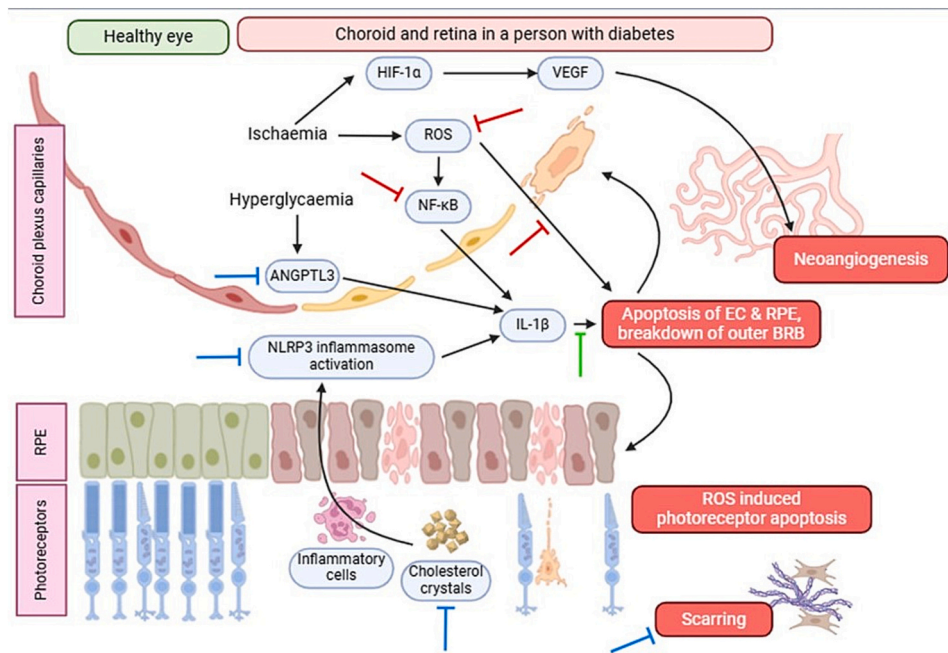


Fig. 1. Potential protective effects of fenofibrate on diabetic retinopathy. Via activation of PPAR α (red inhibitor line) it suppresses ROS generation, inhibits ROS-mediated apoptotic effects, and inhibits NF- κ B activation. Fenofibric acid acts through AMPK-dependent mechanism (green inhibitor line) to suppress IL-1 β mediated apoptosis. Finally, via so far unknown mechanisms (blue inhibitor line) fenofibrate dissolves cholesterol crystals, inhibits NLRP3 inflammasome activation, ANGPTL3 expression and scarring. Image created in BioRender.com.

ANGPTL3: angiopoietin like protein 3; BRB: blood retinal barrier, EC: epithelial cell, HIF-1 α : Hypoxia inducible factor-1 α ; IL-1 β : interleukin 1 β ; NF- κ B: nuclear factor kappa B; NLRP3: NOD-, LRR- and pyrin domain-containing protein 3, PPAR α : peroxisome proliferator activator receptor- α , ROS: reactive oxygen species; RPE: retinal pigmented epithelium; VEGF: vascular endothelial growth factor.

outer BRB, setting up a vicious cycle.²⁹ Intra-retinal accumulation of modified LDL precedes clinical DR, and parallels disease severity as it progresses.⁴⁰ In vitro experiments on human retinal pigment epithelial cells have shown that fenofibric acid may provide protection against modified LDL-driven outer-BRB injury via PPAR α -independent, adenosine monophosphate-activated protein kinase (AMPK)-dependent mechanisms.³⁸ Fenofibrate, as well as other PPAR α agonists like gemfibrozil and bezafibrate (both non-prodrugs), did not confer the same protection on the BRB. This suggests a unique effect of fenofibric acid (the active form released after fenofibrate is cleaved by elastases), possibly mediated through AMPK activation.³⁹

Fenofibrate and its active compound, fenofibric acid, therefore, appear to exert their beneficial effects on DR via mechanisms that are independent of plasma lipid lowering, and these benefits may be mediated through both PPAR α -dependent and AMPK-dependent mechanisms. More recent animal and in vitro studies have uncovered other effects of fenofibrate such as suppression of angiopoietin-like 3 (ANGPTL3) expression,⁴¹ inhibition of NLRP3 (NOD-, LRR- and pyrin domain-containing protein 3) inflammasome activation,⁴² suppression of oxidative stress-mediated activation of Wnt/ β -catenin pathway,⁴³ dissolution of cholesterol crystal deposits⁴⁴ and suppression of reactive gliosis,⁴² all of which are key mechanisms in DR pathogenesis (Fig. 1).

3. Safety

3.1. Effect on renal function

Fenofibrate is primarily excreted renally. A reduced dose of fenofibrate should be offered to patients with impaired renal functions (e.g.: GFR 30–59 mL/min/1.73 m²). Data on safety of fenofibrate among individuals with an eGFR <30 mL/min/1.73 m² are lacking, so it may be best avoided in this group of patients.

The FIELD trial excluded individuals with serum creatinine of 130 μ mol/L or greater. Among recruited participants, an increase in serum

creatinine (by about 10–12 μ mol/L) was observed in participants who received fenofibrate compared to placebo. This difference was already apparent at 6 weeks, and it persisted throughout the follow up (average duration 5 years). However, the rate of eGFR decline was slower in the fenofibrate group compared to placebo group (1.19 vs 2.03 mL/min/1.73 m² per year, $p < 0.001$).⁴⁵ After 8 week treatment wash-out period, creatinine returned towards baseline and, participants in the fenofibrate group had a lesser eGFR reduction from baseline, compared to those in the placebo group (decline from baseline: 1.9 mL/min/1.73 m², $p = 0.065$ in fenofibrate group; 6.9 1.9 mL/min/1.73 m², $p < 0.001$ in placebo group).^{18,45} Furthermore, fenofibrate group experienced a greater reduction in urine albumin to creatinine ratio from baseline compared to placebo group (percentage reduction 24 % vs 11 %, mean difference 14 % [95 % CI 9–18]; $p < 0.001$).⁴⁵ After five years of treatment, the proportion of participants progressing to end-stage kidney disease was not different between fenofibrate and placebo groups (21/4895 vs. 26/4900, $p = 0.48$).⁴⁵

The ACCORD trial excluded individuals with serum creatinine of 133 μ mol/L or greater.⁹ An on-drug/off-drug ancillary study to ACCORD-Lipid trial showed that participants who experience a creatinine rise of >20 % from baseline while on fenofibrate therapy ($n = 321$), return their creatinine to a level comparable to those who received placebo ($n = 565$), 51 days after treatment discontinuation (creatinine in mg/dL: 0.97 ± 0.02 vs. 0.99 ± 0.01).⁴⁶ In contrast, those who experienced less than a 2 % rise of creatinine while on fenofibrate, achieved a lower creatinine concentration than those who received placebo, at 51 days after treatment cessation (creatinine in mg/dL: 0.90 ± 0.02 vs. 0.99 ± 0.01). Among the overall ACCORD trial participants, renal or cardiovascular outcomes were not different between those who experience >20 % rise in creatinine and those who did not.⁴⁶

The LENS trial recruited individuals with an eGFR of >40 mL/min/1.73 m² (allowing for anticipated eGFR decline in the run-in phase) and those with eGFR above 30 mL/min/1.73 m² after the run-in were randomised to receive study drug or placebo. Participants with eGFR

>60 mL/min/1.73 m² received nanoparticle fenofibrate 145 mg or placebo daily, but those with an eGFR of 30–59 mL/min/1.73 m² at the time of randomisation were advised to take 145 mg or placebo on alternate days, and the same benefit on DR progression was observed in this group of patients as with those having daily therapy. Among LENS trial participants, those on fenofibrate had lower trial-averaged eGFRs compared to those on placebo, as expected (difference between groups: −7.9 [−9.1 to −6.8] mL/min/1.73 m²).

The mechanism of the rise in creatinine is unclear. The on-drug/off-drug ancillary study to ACCORD-LIPID trial showed that the fenofibrate-induced increase in serum creatinine and reduction in eGFR paralleled an increase in serum cystatin level, suggesting the creatinine and eGFR trends represent true changes in glomerular filtration.⁴⁷ This may be a result of glomerular hypoperfusion mediated through PPAR α -driven down-regulation of vasodilatory prostaglandin synthesis.⁴⁸

These observations suggest that the rise in serum creatinine with fenofibrate observed as early as 6 weeks into treatment in some individuals is reversible within 6 weeks of treatment discontinuation. The creatinine rise remains reversible even after 5 years, though use of fenofibrate in this clinical situation is likely to extend considerably longer than this for many patients. It is recommended that patients treated with fenofibrate have periodic monitoring of renal function.⁴⁹

3.2. Serious adverse events

Fenofibrate is well tolerated by most patients. Although fibrates have been associated with muscle injury and, rarely, with rhabdomyolysis, these effects were largely observed with gemfibrozil (not in common use now) in combination with the use of statins.⁵⁰ In the FIELD trial 3/4895 in the fenofibrate group and none (0/4900) in the placebo group developed rhabdomyolysis (RR 3.00, 95 % CI 0.3–28.8). Given the rarity of the complication and wide confidence interval, it is difficult to derive a definitive conclusion about the risk of rhabdomyolysis in people taking fenofibrate.

Table 2

Clinical trials evaluating fibrates other than fenofibrate in the treatment of diabetic retinopathy.

Study	Design	Population	Intervention & control	Main DR-related outcome	Remarks
Duncan, 1968 ⁴⁷	Randomised, assessor blinded, controlled clinical trial	Adults with background DR (N = 48 participants)	Atromid 750 mg TDS (n = 23) vs no intervention (n = 25) Minimum follow up 12 months	Primary outcome: Proportion of patients with reduction in exudates: 15/23 (65 %) in atromid group vs 5/25 (20 %) in control group (Chi ² = 13.48, p < 0.0001)	
Nolan, 1969 ⁴⁹	Continuation of Duncan 1968	Adults with background DR (n = 176)	Atromid 750 mg TDS (n = 53) vs no intervention (n = 35) Minimum follow up 12 months	Proportion of eyes with reduction in exudates: 52/106 (49 %) in atromid group vs 10/70 (14 %) in control group	Visual acuity improvement by 2 more lines in Snellen chart: 17 % in atromid group vs 3 % in control group (significance not reported)
Harrold, 1969 ⁴⁸	Randomised double blind placebo-controlled clinical trial	Adults with exudative DR (N = 56 participants)	Clofibrate 1600 mg daily (49 eyes) vs placebo (52 eyes) Follow up 12 months	Primary outcome: Hard exudates improved in treatment group (21 of 44 affected eyes vs 7 of 47 affected eyes, p < 0.001). No significant change in haemorrhages, neovascularisation, visual acuity.	Retinopathy grading based on comparison with standard set of photos.
Emmerich, 2009 ²⁸	Randomised, double-blind, placebo-controlled clinical trial	Adults with T2D and DR (N = 296 participants)	Etofibrate 1 g/day (n = 148) vs placebo (n = 148) Follow up 12 months	Primary outcome: Improvement of ocular pathology based on visual acuity and expert fundoscopic assessment: 46 % in etofibrate group vs 32 % in placebo group, p < 0.001	Benefits apparent within 6 months. High rates of retinopathy improvement in placebo arm compared to other studies.
PROMINENT (Pradhan, 2022) ⁵⁰	Randomised double blind placebo-controlled clinical trial	Adults with T2D with or without DR (N = 10,497), median follow-up 3.4 years	Pemafibrate 0.2 mg twice daily vs placebo	Pre-specified tertiary end point: need for laser therapy or anti-VEGF injection or vitrectomy: 0.86 per 100-person-years vs 0.98 per 100-person-years, HR 0.87, 95 % CI 0.69–1.10	Trial was discontinued prematurely due to cardiovascular futility (primary outcome: CV death, no-fatal MI, ischaemic stroke, coronary revascularisation: 572/5240 [10.9 %] vs 550/5257 [8.8 %], HR 1.03; 95 % confidence interval, 0.91 to 1.15)

CV: cardiovascular, DR: diabetic retinopathy, HR: hazard ratio, T2D: type 2 diabetes, MI: myocardial infarction, VEGF: vascular endothelial growth factor.

Among FIELD study participants, increased risks of pulmonary embolism (0.7 % vs 1.1 %, p = 0.022) and pancreatitis (0.5 % vs 0.8 %, p = 0.031) were observed in analyses not adjusted for multiple statistical testing.¹⁸

In the LENS study, the rates of serious adverse events did not differ between those assigned fenofibrate and placebo.

4. Role of other fibrates

Several RCTs have suggested that clofibrate may decrease the exudate burden in DR^{51,52} (Table 2). However, they did not show consistent decreases in other types of retinal lesions.⁵³ An improvement in visual acuity (by 2 or more levels in Snellen chart or near-vision assessment) was observed in the 5-year follow-up of the participants in one of these trials (18/106 (17 %) in the clofibrate group vs 2/70 (3 %) in the control group).⁵³ However, clofibrate is no longer in use due to its gastrointestinal side effects. One RCT reported that etofibrate (1 g a day) improved DR (determined by fundoscopy and visual acuity assessment) compared to placebo (Table 2).²⁹ However, DR improvement rate was higher in the placebo arm, compared to most other previous studies with fibrates. Differences in population characteristics and methods of measuring outcomes may have contributed to this difference. Ophthalmic outcomes were a tertiary endpoint in the PROMINENT trial (NCT03071692), which investigated the effect of pemafibrate compared to placebo on cardiovascular outcomes, but this study was prematurely discontinued due to cardiovascular futility.⁵⁴ At the time of its discontinuation, no difference was observed in the composite outcome of need for laser treatment, anti-VEGF injections or vitrectomy.

5. Implications on clinical practice and future perspectives

Fenofibrate is currently licensed for treatment of DR in people with T2D in Australia.⁵⁵ Significant reductions of DR progression observed in the LENS trial, the first RCT designed to investigate fenofibrate's effect

on DR progression in participants with non-referable DR, alongside its reasonable safety profile makes fenofibrate a useful and potentially cost-effective adjunct to the treatment of early DR. This calls for reappraisal of guidelines on the management of DR.

There are three further on-going clinical trials directly evaluating the effect of fibrates in DR (Table 3). The FAME-1 trial (NCT01320345) is recruiting individuals with T1D and established early DR. The DRCR study (NCT04661358) in the USA is evaluating fenofibrate in prevention of DR progression in people living with T1D or T2D and the FORTE trial (ACTRN12618000592246) is investigating the role of fenofibrate in reducing central macular thickness among individuals with T2D and DMO. Results from these trials are expected in the coming years.

The long-term effects of fenofibrate therapy on DR remain to be conclusively established. The ACCORDION-EYE study, which involved 1310 participants from the ACCORD-EYE study who were re-assessed for DR progression at 8 years from randomisation, failed to show persistence of the on-treatment protective effects of fenofibrate after its discontinuation at the end of the trial (proportion of participants who progressed by 3 or more steps in ETDRS scale) (11.8 % in fenofibrate group vs 10.2 % in placebo group, aOR 1.13, 95 % CI 0.71–1.79, $p = 0.60$),⁵⁶ suggesting that benefits may not persist after treatment cessation. Therefore, it is likely that fenofibrate may need to be used long term for sustained retinal benefits, similar to statin treatment and blood pressure lowering to reduce CVD risk. Participants in the LENS trial will be followed up for 10 years from the end of the trial.

Fenofibrate appears to be effective irrespective of whether it is used as monotherapy or in combination with a statin. For example, FIELD trial recruited individuals who were not on any other lipid lowering treatment. During the trial, 17 % in the placebo arm and 8 % in the fenofibrate arm went on to commence a statin. It therefore appears highly likely that fenofibrate is effective as a treatment for DR without statin therapy. In the ACCORD-LIPID study, participants with dyslipidaemia were randomised to receive simvastatin and placebo or simvastatin and fenofibrate. In the ACCORD-EYE sub study, participants in the simvastatin and fenofibrate group had improved ocular outcomes. In addition, 75 % of the LENS trial participants were on a statin at baseline. Given that both ACCORD-EYE and LENS trials showed retinopathy benefit, it also appears highly likely that fenofibrate offers retinopathy benefit when added to a statin. There were no significant safety concerns in these studies when fenofibrate was used in combination with statins.

Mounting evidence that fenofibrate provides benefit by means of its actions within the eye suggests that it may prove effective in other retinal diseases beyond DR. For example, studies in mouse models of age-related macular degeneration have shown that fenofibrate inhibits retinal inflammation, oedema and neovascularisation. These appear to result from PPAR α activation,^{37,57} suppression of iron-mediated activation of oxidative stress, and modulation of Wnt/ β -catenin pathway⁵⁸ all of which play key roles in the pathogenesis of the condition.

In conclusion, compelling evidence arising from cardiovascular outcomes trials that have now been supported by a dedicated randomised controlled trial suggest that fenofibrate could slow down the progression of DR over 4–5 years of follow-up. Benefit appears to be comparable between people living with T1D or T2D. It is independent of the lipid lowering effects of fenofibrate and seems likely to be due to its direct actions within the retina. Fenofibrate has been shown to have a good safety profile in the short to medium-term (4–5 years). The decline in renal function appears to be reversible. On-going trials investigating the benefit of fenofibrate in more selected populations (e.g.: individuals with T1D, patients with DMO) will add to this body of evidence. These findings are expected to inform future guidelines and change the clinical management of DR.

CRedit authorship contribution statement

Harsha A. Dissanayake: Writing – original draft, Visualization, Investigation. **Christine A. Kiire:** Writing – review & editing,

Table 3

Ongoing randomised clinical trials evaluating fenofibrate in the treatment of diabetic retinopathy.

Study	Population	Intervention	Outcomes
NCT01320345 (FAME-1 EYE trial)	Adults with T1D and NPDR Target sample size 450	Fenofibrate 145 mg daily vs inert lactose placebo	DR progression (composite of 2-step ETDRS progression, CSMO, need for laser therapy, anti-VEGF injection or vitrectomy)
NCT04661358 (DRCR trial)	Adults with T1D or T2D and mild to moderate NPDR and no CI-DMO. Target sample size 560 Anticipated follow up 6 yrs	Fenofibrate 160 mg daily vs placebo	Composite of progression of DR (2 steps or more in ETDRS or development of PDR) or need for intraocular therapy (laser, injections, vitrectomy)
ACTRN12618000592246 (FORTE trial)	Adults with T2D and DMO Target sample size 392	Fenofibrate 145 mg daily vs placebo for 2 years	Reduction in DMO (measures as central macular thickness)

CI-DMO: centre-involving diabetic macular oedema, CSMO: clinically significant macular oedema, DMO: diabetic macular oedema, DR: diabetic retinopathy, NPDR: non-proliferative diabetic retinopathy, T1D: type 1 diabetes, T2D: type 2 diabetes.

Supervision. **David Preiss:** Writing – review & editing, Supervision. **Garry D. Tan:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

David Preiss is Chief Investigator of the LENS trial for which Mylan have provided study treatment (fenofibrate and retinopathy) free of charge.

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